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Section $\rightarrow$ 'K' (CSE)

Course $\rightarrow$ MAS001

Assignment $\rightarrow$ 5

Section A

A1) means

A2) The null hypothesis in a t-test typically states that there is no significant diff b/w the means of two groups being compared

A3) variances

A4) ANOVA helps to test whether there are significant differences among the means of three or more groups

A5) paired

A6) The null hypothesis in a paired sample t-test is that the mean diff b/w paired observations is zero

A7) is not zero

A8) For a paired sample t-test with n pairs, the df is $n-1$

A9) ongoing

A) The f -ratio or F -test is calculated as the ratio of the variance between groups to the variance within groups. Specifically, it is the mean squared between groups divided by the mean square within groups.

Section 18

B1) $\alpha = 0.05$
 $\bar{x} = 158$
 $\mu_0 = 162.5$
 $s^2 = 39.0625$
 $SE = \sqrt{39.0625} = 6.25 \text{ cm}$
 $n = 10$

$$t = \frac{\bar{x} - \mu_0}{SE/\sqrt{n}} = \frac{158 - 162.5}{6.25/\sqrt{10}} = -2.27$$

$$df = n - 1 = 9$$

for one tailed test with $\alpha = 0.05$
 and $df = 9$, $t_{crit} = -1.833$

$$t_{stat} (-2.275) < t_{crit} (-1.833)$$

$\therefore$ we reject null hypothesis.

B2) Battery X, $n_1 = 10$, $\mu_{mean} = 1000 \text{ hours}$
 $\sigma_{variance} = 600$

Battery Y, $n_2 = 12$, $\mu_{mean} = 2000 \text{ hours}$,
 $\sigma_{variance} = 121$

$$t = \frac{\mu_1 - \mu_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{1000 - 2000}{\sqrt{\frac{600^2}{10} + \frac{121^2}{12}}} = -2.231$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left(\frac{s_1^4}{n_1^2} + \frac{s_2^4}{n_2^2}\right)} = 19.8$$

for two tailed test at 5% significance
 $df = 19$, the critical t -value is ± 2.033

$$t_{stat} -2.231 > 2.033$$

$\therefore$ we reject null hypothesis

$\therefore$ there is a significant diff between two battery types

B3) city 1, $n_1 = 13$, $S_1 = \$27.50$
 city 2, $n_2 = 16$, $S_2 = \$25.75$
 $\alpha = 0.01$

variance $S_1^2 = (27.50)^2 = 756.25$
 $S_2^2 = (25.75)^2 = 663.0625$

f-statistic:

$$f = S_1^2 / S_2^2 = \frac{756.25}{663.0625} = 1.17$$

DoF, $df_1 = n_1 - 1 = 13 - 1 = 12$
 $df_2 = n_2 - 1 = 16 - 1 = 15$

for two tailed test at $\alpha = 0.01$
 $\alpha/2 = 0.005$ in upper tail

critical t-value for $df_1 = 12$
 $df_2 = 15$ at $\alpha/2 = 0.005$
 expected to be significantly higher
 from the calculated t-stat of 1.17

t-stat (1.17) < critical

$\therefore$ fail to reject the null hypothesis

sample proportions

total females, $p_1 = \frac{1236}{1800} = 0.6867$

non-Hispanic females

total non-Hispanic females = 2000 - 1800 = 200

prob for common among non-Hispanic
 $= 1400 / 2000 = 0.7$

$p_2 = 1400 / 2000 = 0.7$

pooled proportion $\Rightarrow p = \frac{1400}{2000} = 0.7$

$SE = \sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$

$SE = \sqrt{0.7(1-0.7) \left(\frac{1}{1800} + \frac{1}{200} \right)} = 0.0342$

$z = \frac{p_1 - p_2}{SE} = \frac{0.6867 - 0.7}{0.0342} = \frac{-0.0133}{0.0342} = -0.39$

two tailed test at 5% level of significance ($\alpha = 0.05$)

critical 2 values are ± 1.96

Computed $z = -3.90$ lies outside
the range -1.96 to 1.96

$\therefore$ we reject the null hypothesis

B5) Mean calculations

$$\text{Food I} = (8+13+19+8+10)/5 = 10.6$$

$$\text{Food II} = (4+5+4+6+9)/5 = 5.6$$

$$\text{Food III} = (11+8+7+13+7)/5 = 9.2$$

$$\text{Overall mean} = (33+28+46)/15 = 8.4667$$

Sum of squares b/w groups (SSA)

$$\begin{aligned} \text{SSB} &= 5 \times (10.6 - 8.4667)^2 + 5 \times (5.6 - 8.4667)^2 \\ &\quad + 5 \times (9.2 - 8.4667)^2 \\ &= 66.555 \end{aligned}$$

Sum of squares within groups (SSW)

$$\text{Food I} = 6.76 + 1.96 + 70.56 + 8.76 + 2.16 = 90.16$$

$$\text{Food II} = 2.56 + \dots + 11.56 = 17.2$$

$$\text{Food III} = 3.24 + 1.44 + 4.84 + \dots + 4.84 = 28.8$$

$$\text{Total SSW} = 107.2 + 17.2 + 28.8 = 153.2$$

Mean squares

$$\text{MSB} = 66.555 / (3-1) = 33.2775$$

$$\text{MSW} = 153.2 / (15-3) = 12.7667$$

$$F_{\text{stat}} = \frac{\text{MSB}}{\text{MSW}} = \frac{33.2775}{12.7667} = 2.608$$

for $\text{df}_1 = 2$, $\text{df}_2 = 12$ at $\alpha = 0.05$

critical value = 3.885

$\because 2.608 < 3.885$ $\therefore$ reject null hypothesis

Solution - C

$$\begin{aligned} \text{C1) } \bar{x} &= \frac{56+49+44+52+45+48+49+41+49+41}{10} \\ &= 47.3 \end{aligned}$$

Devi from mean = 2.7, 1.7, -3.3, 4.7, -2.3,
0.2, -1.3, -2.3,

$$\text{S2 dev} = 7.29 + 2.89 + 10.89 + 22.09 + \dots + 5.29$$

$$\text{Sum of 12 dev} = 64.1$$

Sample variance $S^2 = \frac{64.1}{9} \approx 7.1222$

$S = \sqrt{7.1222} = 2.668$

$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{47.3 - 50}{2.668/\sqrt{10}}$

$= -3.1\%$

df: $n-1 = 9$

for $\alpha = 0.05$, critical value $= \pm 2.262$

$-3.199 < -2.262$

$\therefore$ reject null hypothesis

(2) Group 1 mean = $\frac{18+20+36+50+49}{5} = \frac{173}{5} = 34.6$

Group 2 mean = $\frac{29+28+26+35+30}{5} = \frac{148}{5} = 29.6$

Group 1 variance $S_1^2 = \frac{(18-34.6)^2 + (20-34.6)^2 + \dots + (49-34.6)^2}{9}$

$= \frac{141.75}{9} = 15.75$

Group 2 variance $S_2^2 = \frac{(29-29.6)^2 + (28-29.6)^2 + \dots + (30-29.6)^2}{4}$

$= \frac{14.33}{4} = 3.58$

$t_{stat} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$

$= \frac{34.6 - 29.6}{\sqrt{\frac{15.75}{5} + \frac{3.58}{5}}} = \frac{5}{\sqrt{4.66}} = 2.34$

$= 0.001$

dot = $\frac{(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2})^2}{(\frac{S_1^2}{n_1})^2 + (\frac{S_2^2}{n_2})^2}$

dot = $\frac{(15.75 + 3.58)^2}{\frac{15.75^2}{5} + \frac{3.58^2}{5}} \approx 0.13$

for two tailed test with $df = 13$ and $\alpha = 0.05$ the critical value is 2.160

calculated t stats is 0.601 < 2.160

p value for $t = 0.601$ with $df = 13$ is 0.555 > 0.005

$\therefore$ t stats < critical and p value > significance level

$\therefore$ we fail to reject null hypothesis

(3) $\alpha = 0.10$

mean A = 64 + 67 = 131 / 2 = 65.5

mean B = 64 + 66 = 130 / 2 = 65

$$S_A^2 = (66 - 65.5)^2 + (67 - 65.5)^2 = 1.5 / 2 = 0.75$$

$$S_B^2 = (64 - 65)^2 + (66 - 65)^2 = 2 / 2 = 1$$

$$F = \frac{S_A^2}{S_B^2} = \frac{0.75}{1} = 0.75$$

for $\alpha = 0.10$ (two tailed test)
 $df_1 = 10$ $df_2 = 8$

$F_{\text{actual}} = 0.75$ $F_{\text{table}} = 3.60$

the F value is less than F_{table}

$F = 0.75$ less than $F_{\text{table}} = 3.60$, $F_{\text{table}} = 3.60$

$\therefore$ falls in acceptance region

(c) test statistic for chi-square test

$$\chi^2 = (O - E)^2 / E$$

O = observed frequency E = expected frequency

E = Row total $\times$ Column total / Grand total

$$df = (r - 1) \times (c - 1)$$

calculating χ^2

	Lowest school	Public school	Total
For car	$800 \times 500 / 1000 = 400$	$800 \times 500 / 1000 = 400$	800
Low			
High	$200 \times 500 / 1000 = 100$	$200 \times 500 / 1000 = 100$	200
Total	500	500	1000

$$\chi^2 = \frac{(370-400)^2}{400} + \frac{(420-400)^2}{400} + \frac{(130-400)^2}{100} + \frac{(20-100)^2}{100}$$

$$\chi^2 = 22.5$$

$$df = (2-1) \times (2-1) = 1$$

$$\alpha = 0.05 \text{ and } df = 1$$

$$\chi^2_{\text{critical}} = 3.841$$

$$\because \chi^2 (22.5) > \chi^2_{\text{critical}} (3.841)$$

$\therefore$ we reject H_0

$$(5) \bar{x}_{\text{medium}} = \frac{10+12+9+11+13}{5} = 11.2$$

$$\bar{x}_{\text{exam}} = \frac{6+8+3+10+2}{5} = 3.8$$

$$\bar{x}_{\text{dist}} = \frac{5+5+12+8+4}{5} = 7.6$$

$$\bar{x}_{\text{total}} = \frac{(10+12+9+11+13) + (6+8+3+10+2) + (5+5+12+8+4)}{15} = 8$$

$$\begin{aligned} \text{Total Sum of Squares} &= \sum_{i=1}^n (x_i - \bar{x}_{\text{total}})^2 \\ &= (10-8)^2 + (12-8)^2 + \dots \\ &= 129.2 \end{aligned}$$

$$SST_{\text{medium}} = (10-8)^2 + \dots + (13-8)^2 = 95$$

$$SST_{\text{exam}} = (6-8)^2 + (8-8)^2 + \dots + (2-8)^2 = 129$$

$$SST_{\text{dist}} = (5-8)^2 + \dots + (4-8)^2 = 42$$

$$SST = 95 + 129 + 42 = 266$$

Between group sum of SS (SSB)

$$= \sum_{j=1}^k n_j (\bar{x}_j - \bar{x}_{\text{total}})^2$$

$$SSB_{\text{medium}} = 5(11.2-8)^2 = 72.2$$

$$SSB_{\text{exam}} = 5(3.8-8)^2 = 88.2$$

$$dist = 5(7.6-8)^2 = 0.8$$

$$SSB = 72.2 + 88.2 + 0.8 = 161.2$$

Within group sum of squares

$$SSW = SST - SSB = 266 - 161.2 = 104.8$$

$$\text{B/w group mean sq} = \frac{SSB}{k-1} = \frac{161.2}{3-1} = 80.6$$

$$\begin{aligned} \text{Within-groups mean sq} &= \frac{SSW}{n-k} = \frac{104.8}{15-3} \\ &= 8.73 \end{aligned}$$

$$F_{stat} = \frac{MSB}{MSW} = \frac{80.8}{8.73} \approx 9.23$$

$$df_1 = 4-1 = 3-1 = 2$$

$$df_2 = N-4 = 17-3 = 12$$

at 5% significance level ($\alpha = 0.05$)
using t-table
critical value (2, 12) = 3.22

$t > t_{critical}$

$\therefore$ Reject null hypothesis

Section - D

D1) test statistic for one sample t-test

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$\bar{x} = 4.2 + 4.6 + 4.1 + \dots + 5.0 / 10$$

$$= 4.4$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{(4.2-4.4)^2 + (4.6-4.4)^2 + \dots + (5.0-4.4)^2}{10-1}}$$

$$= 0.523$$

$$t = \frac{4.4 - 4.0}{0.523/\sqrt{10}} = 2.15$$

$$df = n-1 = 10-1 = 9$$

$\alpha = 0.05$ for two tailed test

critical value $\alpha = 0.05$ and $df = 9 = \pm 2.262$

or $|t| = 2.15 < 2.262$ $\therefore$ we fail to reject null hypothesis

$$D2) \bar{x}_A = 10 + 6 + 10 + 11 = 37 / 4 = 9.25$$

$$\bar{x}_B = 7 + 13 + 22 + 1 = 43 / 4 = 10.75$$

$$s_A^2 = \frac{\sum (x_{Ai} - \bar{x}_A)^2}{n_A - 1}$$

$$= \frac{(10-9.25)^2 + (6-9.25)^2 + (10-9.25)^2 + (11-9.25)^2}{4-1}$$

$$= 18.33$$

$$s_B^2 = \frac{(7-10.75)^2 + (13-10.75)^2 + (22-10.75)^2 + (1-10.75)^2}{4-1}$$

$$= 28.55$$

for pooled standard deviation

$$s_p = \sqrt{(n_A-1)s_A^2 + (n_B-1)s_B^2 / (n_A + n_B - 2)}$$

$$SP = \frac{(10+1)(13.33) + (12+1)(12.55)}{10+12-2} = 4.66$$

$$t = \frac{\bar{x}_A - \bar{x}_B}{SP \times \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} = \frac{12-15}{4.66 \times \sqrt{\frac{1}{10} + \frac{1}{12}}} = -1.50$$

$$df: n_A + n_B - 2 = 10 + 12 - 2 = 20$$

$$\alpha = 0.05$$

$$\text{from table } df = 20, \alpha = 0.05$$

$$t_{critical} = \pm 2.086$$

$$\text{value } (-1.50) \text{ lies in } -2.086 \leq t \leq 2.086$$

we fail to reject null hypothesis

Q3)

$$\bar{x} = 8260 + 8130 + \dots + 8340 / 5 = 8230$$

$$S_1^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$= \frac{(8260 - 8230)^2 + \dots + (8340 - 8230)^2}{5-1} = 15750$$

$$n^2 = 7950 + 9950 + \dots / 6 = 7956.67$$

$$S_2^2 = \frac{(7950 - 7956.67) + \dots}{6-1} = 11369.08$$

$$f_1 = \frac{S_1^2/S_2^2}{11369.08} = 1.305$$

$$df_1 = n_1 - 1 = 5 - 1 = 4, df_2 = n_2 - 1 = 5$$

$$\text{upper critical value } f_{0.025} = 6.956$$

$$\text{lower " } f_{0.025} = 0.160$$

value (1.305) lies between critical values

$\therefore$ we fail to reject null hypothesis

Q4)

$$\bar{x}_1 = 2+3+5+0+6/5 = 3.6$$

$$\bar{x}_2 = 4+6+8+4+9+0+6/7 = 4.71$$

$$\bar{x}_3 = 5+2+3+2+3+3/6 = 3.0$$

$$\text{overall mean } \bar{\bar{x}} = \frac{2+3+5+\dots+17}{18} = 3.83$$

$$\text{Total sum of squares } SST = \sum (n_i - \bar{\bar{x}})^2 = 12 - 3.83^2 + (3 - 3.83)^2 + \dots = 36.62$$

Between group sum of 12 SS

$$= n_1 (\bar{x}_1 - \bar{\bar{x}})^2 + n_2 (\bar{x}_2 - \bar{\bar{x}})^2 + \dots$$

$$= 5(3.6 - 3.82)^2 + 7(4.71 - 3.82)^2$$

$$= 9.83$$

Within group of squares $SSW = SST - SSB$

$$= 36.67 - 9.83$$

$$= 26.84$$

$$df_B = k - 1 = 3 - 1 = 2$$

$$df_W = n - k = 18 - 3 = 15$$

$$df_T = N - 1 = 18 - 1 = 17$$

Mean Squares

$$MSB = \frac{SSB}{df_B} = \frac{9.83}{2} = 4.92$$

$$MSW = \frac{SSW}{df_W} = \frac{26.84}{15} = 1.79$$

$$F = \frac{MSB}{MSW} = \frac{4.92}{1.79} = 2.75$$

By F distribution $df_B = 2$ and $df_W = 15$ at
1% significance level

$$F_{0.01, 2, 15} = 6.36$$

F value (2.75) < $F_{critical}$ (6.36)

we fail to reject the null hypothesis